

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 5: Acids, Bases and Salts

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	5 — Acids, Bases and Salts
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Paper 3 (first hard tier). Traps: confusing the colour an indicator shows with the family it names; assuming "no litmus change" proves a substance is water; treating neutralisation as making a mixture acidic or basic rather than neutral; mixing up which member of a neutralisation pair to add to a sting, soil or waste; reading a single indicator result as conclusive when a second indicator contradicts it; thinking strength of taste equals chemical family. Reason in 2-4 steps; no recall gimmies. Do not write on this paper — mark answers on your answer sheet.

1. A juice tastes sharply sour. Before any test is done, the most reliable prediction you can make about how it will behave with litmus is that it will:

- (A) turn red litmus blue, because sour things are bases
- (B) leave both litmus papers unchanged, because taste tells nothing about litmus
- (C) turn blue litmus red, because sourness is a property of acids
- (D) turn turmeric paper red, because sour things make turmeric red

2. Two unlabelled bottles are tested. Bottle 1 turns red litmus blue; bottle 2 turns blue litmus red. If you accidentally mix roughly equal amounts of the two, the litmus result of the mixture is most likely:

- (A) no change in either litmus, because an acid and a base tend to neutralise each other
- (B) turns blue litmus red, because the acid is always stronger
- (C) turns red litmus blue, because the base is always stronger
- (D) turns both litmus papers green, because mixing produces a new colour

3. A student says: "Milk of magnesia must be an acid because people take it for a stomach problem." The correct correction is:

- (A) it is a base; it is taken precisely to neutralise the excess acid causing acidity
- (B) it is an acid; that is why it cures the acidic stomach
- (C) it is neutral, so it simply dilutes the stomach acid
- (D) it is a salt that washes the acid away without reacting

4. Phenolphthalein is added to four colourless liquids: lemon juice, distilled water, sugar water, and lime water. The number of these in which it turns pink is:

- (A) 2
- (B) 3
- (C) 0
- (D) 1

5. A drop of china rose (hibiscus) indicator is added to vinegar. Knowing china rose turns dark pink (magenta) in acids and green in bases, the colour you should expect is:

- (A) green, because vinegar is acidic
- (B) dark pink (magenta), because vinegar is acidic
- (C) green, because vinegar is basic
- (D) colourless, because vinegar is neutral

6. When dilute hydrochloric acid is exactly neutralised by sodium hydroxide, then the water is evaporated away, what is left behind is:

- (A) sodium hydroxide left over from the base
- (B) hydrochloric acid that did not react
- (C) common salt (sodium chloride)
- (D) a fresh acid stronger than the first

7. An ant's sting causes a burning pain because it injects formic acid. A friend suggests rubbing lemon juice on it. The best judgement is:

- (A) yes, lemon juice is a base and will neutralise the acid sting
- (B) yes, adding more acid cancels the first acid
- (C) no, only water can ever relieve a sting
- (D) no, lemon juice is acidic and will not relieve an acid sting; a base such as baking soda paste is needed

8. A liquid leaves both red and blue litmus unchanged. A classmate concludes "it must be pure water." The most accurate response is:

- (A) not necessarily; any neutral substance, such as salt water or sugar water, also leaves litmus unchanged
- (B) correct, only pure water leaves both litmus papers unchanged
- (C) wrong, because water actually turns blue litmus red
- (D) wrong, because all neutral liquids turn red litmus blue

9. A factory's waste water turns blue litmus red and must be treated before it enters a river. The correct treatment is to add:

- (A) more acid, to balance the waste
- (B) a base such as slaked lime, to neutralise the acidic waste
- (C) common salt, to soak up the acid
- (D) sugar, to sweeten and neutralise it

10. Turmeric paper is touched with a drop of a sample and stays yellow. From this single result you can correctly conclude that the sample is:

- (A) definitely an acid
- (B) not a base (it could be an acid or a neutral substance)
- (C) definitely neutral
- (D) definitely a base

11. Which set of three substances lists, in order, one acid, one base, and one neutral substance?

- (A) soap solution, lemon juice, sodium hydroxide
- (B) tamarind water, lime water, sugar solution
- (C) distilled water, vinegar, milk of magnesia
- (D) baking soda solution, curd, common salt water

12. A gardener's soil is too acidic and a neighbour's pond water is too basic for fish. Which single substance could correct the acidic soil but would worsen the basic pond?

- (A) vinegar, which fixes both because it is neutral
- (B) lemon juice, a base that suits both
- (C) slaked lime, a base that fixes acidic soil but adds to basicity in the pond
- (D) sugar, which neutralises acids and bases alike

13. A solution turns red litmus blue. Which other indicator result must it also give?

- (A) it turns turmeric paper red
- (B) it turns blue litmus red
- (C) it leaves phenolphthalein colourless
- (D) it turns china rose dark pink

14. Baking powder is a dry mix of sodium bicarbonate (a base) and a powdered edible acid. It keeps for months in a sealed tin without fizzing, yet makes a cake batter rise as soon as it is stirred in. The reason it stays inert while dry is that:

- (A) the acid in the powder loses its sourness while stored dry
- (B) the acid and base can only react and release their gas once water lets them meet
- (C) water is a base that switches the powder on
- (D) the rising gas is merely air that was beaten into the dry powder

15. Phenolphthalein-pink soap solution is titrated by adding lemon juice drop by drop. The moment the pink colour just vanishes tells you that:

- (A) the solution has become strongly basic
- (B) the lemon juice has turned into a base
- (C) enough acid has been added to neutralise the base, leaving the solution about neutral
- (D) the soap has become an even stronger base

16. Among litmus, turmeric, china rose and phenolphthalein, the one that is NOT extracted from a plant or living source is:

- (A) litmus
- (B) turmeric
- (C) china rose
- (D) phenolphthalein

17. Baking soda and vinegar are mixed inside a sealed sandwich bag, which quickly swells tight and could burst. The swelling is best explained by:

- (A) the vinegar evaporating into vapour because the bag is warm
- (B) air leaking into the bag from outside
- (C) the baking soda merely dissolving and taking up room
- (D) a gas being produced as the base (baking soda) neutralises the acid in vinegar

18. Acid rain slowly eats away marble statues. This happens because:

- (A) marble is acidic and reacts with a base in the rain
- (B) the rain is neutral and simply washes the marble away
- (C) marble is calcium carbonate, a base, which reacts with the acid in the rain
- (D) the marble heats up in the rain and cracks

19. A wasp sting is alkaline (basic), unlike a bee or ant sting. To relieve a wasp sting you should apply something:

- (A) mildly basic, such as baking soda paste
- (B) mildly acidic, such as vinegar
- (C) neutral, such as plain water only
- (D) strongly basic, such as sodium hydroxide

20. Three liquids X, Y, Z are tested. X turns turmeric red, Y turns blue litmus red, Z changes neither litmus. Which liquid will turn phenolphthalein pink?

- (A) X, because turning turmeric red shows it is basic
- (B) Y, because it turned blue litmus red
- (C) Z, because it changed neither litmus
- (D) Y, because phenolphthalein turns pink in any acid

21. A student writes the word equation: acid + base gives salt + water. Which everyday event is the best real-life example of this equation?

- (A) taking an antacid tablet to relieve an acidic stomach
- (B) ice cubes melting in a warm drink
- (C) a cut apple turning brown in air
- (D) salt dissolving in cooking water

22. A sample turns blue litmus red and also leaves phenolphthalein colourless. These two results together tell you the sample is:

- (A) basic, because phenolphthalein is involved
- (B) neutral, because phenolphthalein stayed colourless
- (C) acidic, and the two results agree
- (D) giving contradictory results that cannot both be true

23. Pure water is described as neutral. The clearest single piece of evidence for this is that water:

- (A) turns blue litmus red like an acid
- (B) turns red litmus blue like a base
- (C) turns phenolphthalein pink
- (D) leaves both red and blue litmus papers unchanged

24. An eggshell, which is mostly calcium carbonate, is left overnight in a cup of vinegar; bubbles stream off it and the hard shell gradually disappears. The best explanation is:

- (A) the carbonate shell reacts with the acid, releasing a gas as the shell is eaten away
- (B) the vinegar dries the shell out until it crumbles
- (C) the shell dissolves unchanged, just as sugar dissolves in water
- (D) the egg inside warms the shell until it flakes apart

25. Methyl orange is red in acids and yellow in bases. A drop is added to soap solution. The colour seen is:

- (A) red, because soap solution is acidic
- (B) red, because soap solution is basic
- (C) yellow, because soap solution is basic
- (D) colourless, because soap solution is neutral

26. A chemist neutralises sulphuric acid with sodium hydroxide. Besides water, the product is a salt called sodium sulphate. This shows that the salt formed in a neutralisation:

- (A) is always common salt no matter the acid and base
- (B) is always the same as the starting acid
- (C) is always sodium hydroxide
- (D) depends on which acid and which base were used

27. Lime water is exposed to enough carbon dioxide (an acidic gas) until it is just neutralised. Its effect on litmus then becomes:

- (A) turns red litmus blue, because it is still strongly basic
- (B) no change in either litmus, because the base has been neutralised
- (C) turns blue litmus red, because the gas made it strongly acidic
- (D) turns both litmus papers green

28. Which statement about indicators is the only correct one?

- (A) phenolphthalein is pink in acids and colourless in bases
- (B) blue litmus turns red in a base
- (C) turmeric is yellow in acids and red in bases
- (D) china rose is dark pink in bases and green in acids

29. A solution turns china rose green. The same solution is therefore expected to:

- (A) turn blue litmus red and taste sour
- (B) leave both litmus papers unchanged
- (C) turn turmeric paper stay yellow
- (D) turn red litmus blue and feel soapy

30. A teacher mixes an acid and a base and the beaker becomes noticeably warm. The warmth is best explained as:

- (A) heat released during the neutralisation reaction
- (B) the acid evaporating and cooling the beaker
- (C) the base freezing the mixture
- (D) sunlight heating the glass

31. Among the substances curd, lime water, ammonia solution, and tamarind water, how many are bases?

- (A) 1 (B) 3
(C) 2 (D) 4

32. A label-less bottle holds a liquid that feels soapy and turns turmeric paper red. Without tasting it, the safest classification is:

- (A) an acid (B) a base
(C) a neutral substance (D) common salt solution

33. A river fish-kill is traced to acidic runoff. The cheapest sensible fix recommended is to spread a base over the affected water. Which choice fits both "a base" and "cheap and commonly used"?

- (A) vinegar (B) lemon juice
(C) more of the acidic runoff (D) lime (slaked lime)

34. A student argues: "A stronger sour taste must mean the liquid is a base, because bases are the strong ones." The correct judgement is:

- (A) correct; strong taste always means a base
(B) correct; sourness becomes basic when it is strong
(C) wrong; sourness means neutral, not acid
(D) wrong; sour taste is an acid property regardless of strength, and strength does not switch the family

35. Four samples give these turmeric results: P stays yellow, Q turns red, R stays yellow, S turns red. How many of the four are basic?

- (A) 1 (B) 3
(C) 4 (D) 2

36. Which pairing of a substance with its family is INCORRECT?

- (A) vinegar is an acid (B) lime water is a base
(C) milk of magnesia is an acid (D) curd is an acid

37. Hydrochloric acid in the stomach is useful, but too much causes a burning feeling. An antacid relieves this by:

- (A) adding more acid so the stomach gets used to it
(B) coating the stomach with sugar so no reaction occurs
(C) neutralising the excess acid with a mild base to form salt and water
(D) turning the stomach acid into a stronger acid

38. A drop of phenolphthalein in solution 1 is pink and in solution 2 is colourless. Litmus shows solution 2 turns blue litmus red. Solutions 1 and 2 are, respectively:

- (A) a base and an acid (B) an acid and a base
(C) a base and a neutral substance (D) two bases

39. Lemon juice and baking soda solution are each tested with red litmus, blue litmus, and phenolphthalein. The substance that turns red litmus blue and phenolphthalein pink is:

- (A) lemon juice, because it is a base
- (B) lemon juice, because acids turn red litmus blue
- (C) neither, because both are acids
- (D) baking soda solution, because it is a base

40. Carbonic acid forms when carbon dioxide dissolves in water, such as in a fizzy drink. This tells you that a fizzy soft drink is most likely:

- (A) mildly basic
- (B) mildly acidic
- (C) strongly basic
- (D) perfectly neutral like pure water

41. A solution leaves litmus unchanged but is claimed to be a base. Which result would prove that claim wrong?

- (A) phenolphthalein added to it turns pink
- (B) turmeric added to it turns red
- (C) china rose added to it turns green
- (D) phenolphthalein added to it stays colourless, as expected for a non-base

42. Two students each add lime to a field. Student 1's field was acidic and improves; student 2's field was already basic and gets worse. This best illustrates that lime is:

- (A) an acid, useful for basic soil
- (B) a base, useful for acidic soil but harmful to add to basic soil
- (C) neutral, so it never changes the soil
- (D) a salt that has no effect on soil pH

43. In a neutralisation between an acid and a base, the role of the water-litmus test at the endpoint is to confirm that:

- (A) blue litmus turns red, showing acid is left over
- (B) red litmus turns blue, showing base is left over
- (C) neither litmus changes, showing the mixture has reached neutral
- (D) the mixture has become a fresh acid

44. Which one of these would turn dry turmeric paper red on contact?

- (A) a drop of lemon juice
- (B) a drop of soap solution
- (C) a drop of vinegar
- (D) a drop of sugar water

45. A natural indicator and a synthetic indicator give matching information about acids and bases. Which pair is one natural and one synthetic indicator?

- (A) litmus and turmeric
- (B) china rose and phenolphthalein
- (C) turmeric and china rose
- (D) litmus and china rose

46. A bee sting injects an acid. A child rubs honey on it expecting relief. The best assessment is:

- (A) honey is a base and neutralises the sting
- (B) honey is the perfect cure because it is sweet
- (C) any sweet substance neutralises acids
- (D) honey will not neutralise an acid sting; a mild base like baking soda is needed

47. A solution turns blue litmus red. It is then mixed with a little milk of magnesia and tested again, now turning red litmus blue. The change shows that:

- (A) the milk of magnesia was an acid all along
- (B) the original solution was a base
- (C) enough base was added to overshoot neutral and make the mixture basic
- (D) no reaction happened between them

48. Which of the following correctly explains why distilled water has no effect on phenolphthalein?

- (A) distilled water is neutral, and phenolphthalein only changes colour in a base
- (B) distilled water is acidic, and phenolphthalein needs an acid
- (C) distilled water is basic but too weak to show colour
- (D) phenolphthalein only works with salt water

49. A bottle is found to turn blue litmus red, leave phenolphthalein colourless, and turn china rose dark pink. All three results together indicate the contents are:

- (A) a base, because china rose changed colour
- (B) an acid, since all three results point to acidity
- (C) neutral, because phenolphthalein stayed colourless
- (D) contradictory and impossible to classify

50. Sodium chloride solution (common salt in water) is tested with all four indicators. The combined result is:

- (A) blue litmus red, turmeric red, phenolphthalein pink
- (B) no litmus change, turmeric yellow, china rose its original colour, phenolphthalein colourless
- (C) red litmus blue, turmeric red, phenolphthalein pink
- (D) blue litmus red, turmeric yellow, phenolphthalein pink

51. Acid rain is mainly caused by gases like sulphur dioxide and nitrogen oxides dissolving in rain. The basic monuments most damaged by it are made of:

- (A) glass windows, which are acidic
- (B) iron railings, which are bases
- (C) marble and limestone (calcium carbonate)
- (D) plastic sheets, which neutralise acid

52. A student tests four kitchen liquids with blue litmus: vinegar to red, lime water no change, lemon juice to red, milk no change. The number of acids identified is:

- (A) 2
- (B) 1
- (C) 3
- (D) 4

53. Why can a single indicator result sometimes be misleading, requiring a second test?

- (A) litmus turns the same colour in acids and bases, so it is useless
- (B) turmeric turns red in both acids and bases

(C) every indicator gives the same colour in every solution

(D) phenolphthalein stays colourless in both acids and neutrals, so a colourless result alone cannot tell them apart

54. A glass of milk is roughly neutral when fresh but turns sour as it spoils. The souring shows that:

(A) an acid (lactic acid) has formed, so spoiled milk is acidic

(B) a base has formed, making it basic (C) it has become more neutral than before

(D) salt has formed, making it a salt solution

55. A neutralisation is done with the right amounts so that no acid or base is left. If you wanted to recover the salt, the correct final step is to:

(A) add more acid to make the salt appear (B) add more base to make the salt appear

(C) cool the solution to freeze out the acid

(D) evaporate the water, leaving the solid salt behind

56. A liquid turns red litmus blue. Which everyday substance could it be?

(A) orange juice

(B) vinegar

(C) soap solution

(D) soda water

57. A child mixes lemon juice with baking soda solution little by little until adding more causes no further fizzing and litmus shows no change. At that point the mixture is:

(A) neutral, with the acid and base having neutralised each other

(B) still strongly acidic

(C) still strongly basic

(D) now a pure acid stronger than lemon juice

58. A teacher lists tests of an unknown liquid: it tastes bitter, feels soapy, turns red litmus blue, and turns phenolphthalein pink. The single best conclusion is:

(A) the liquid is an acid, because it feels soapy

(B) the liquid is a base, since all four results are base properties

(C) the liquid is neutral, because so many tests agree

(D) the results conflict, so no conclusion is possible

59. Quicklime added to soil reacts with water and acidity. A farmer wonders whether the same lime would help if the soil were instead too basic. The correct advice is:

(A) no; for basic soil add something acidic, such as decaying compost, not more lime

(B) yes; lime fixes any soil because it is neutral

(C) yes; lime is an acid and neutralises basic soil

(D) no; basic soil should be treated with sodium hydroxide

60. A drop of the same china rose indicator is added to two unknowns: in cup A it turns dark pink (magenta), in cup B it turns green. Which cup, treated with phenolphthalein instead, would turn pink?

- (A) cup B, because green in china rose means it is basic
- (B) cup A, because dark pink means it is basic
- (C) cup A, because phenolphthalein matches dark pink
- (D) neither, because china rose and phenolphthalein disagree

Solutions — Science (Class 7), Chapter 5: Acids, Bases and Salts — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	A	31	C	41	D	51	C
2	A	12	C	22	C	32	B	42	B	52	A
3	A	13	A	23	D	33	D	43	C	53	D
4	D	14	B	24	A	34	D	44	B	54	A
5	B	15	C	25	C	35	D	45	B	55	D
6	C	16	D	26	D	36	C	46	D	56	C
7	D	17	D	27	B	37	C	47	C	57	A
8	A	18	C	28	C	38	A	48	A	58	B
9	B	19	B	29	D	39	D	49	B	59	A
10	B	20	A	30	A	40	B	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Sourness is the hallmark taste of acids, and acids turn blue litmus red, so that is the safe prediction. Turning red litmus blue is a base behaviour, and bases taste bitter, not sour. Saying taste reveals nothing ignores the reliable link between sour taste and acid character. Turmeric turning red is a base reaction, not an acid one; acids leave turmeric yellow.

2. (A) Bottle 1 is a base (red litmus to blue) and bottle 2 is an acid (blue litmus to red). Mixing roughly equal amounts neutralises them, giving a near-neutral mixture that leaves litmus unchanged. There is no rule that the acid or the base is always stronger, so neither one-sided result is justified. Litmus only goes red or blue; green is not a litmus colour.

3. (A) Milk of magnesia (magnesium hydroxide) is a mild base; an antacid relieves acidity because a base neutralises the excess stomach acid. Calling it an acid is the misconception, since adding more acid would worsen acidity. A neutral substance would only dilute, not

neutralise, and would give little relief. A salt does not neutralise an acid the way a base does.

4. (D) Phenolphthalein turns pink only in a base. Of the four, only lime water is basic, so exactly one turns pink. Lemon juice is acidic and sugar water and distilled water are neutral, all of which leave phenolphthalein colourless. So the answer is one, not two, three, or zero.

5. (B) Vinegar contains acetic acid, so it is acidic, and china rose turns dark pink (magenta) in acids. Green is the colour china rose shows in bases, not acids, so green is wrong here. Vinegar is not basic and not neutral, so those reasons fail too.

6. (C) Neutralisation gives salt and water; here HCl plus NaOH gives sodium chloride and water, and evaporating the water leaves the salt. If the acid and base were exactly neutralised, none of the base is left over, so leftover NaOH is wrong. Likewise no unreacted acid remains. Neutralisation produces a salt, not a new stronger acid.

7. (D) The sting is acidic, so relief needs a mild base, and baking soda (sodium bicarbonate) paste neutralises it. Lemon juice is itself acidic, so it cannot neutralise the acid and would not help. Two acids do not cancel each other. Plain water only dilutes, while the proper remedy is a base, so saying only water works is wrong.

8. (A) No litmus change means the liquid is neutral, but many neutral liquids exist besides water, such as salt water and sugar water, so the conclusion is unsafe. Water does not turn blue litmus red, since it is neutral. Neutral liquids do not turn red litmus blue either; that is a base reaction.

9. (B) Blue litmus turning red shows the waste is acidic, so it must be neutralised with a base; slaked lime is the standard cheap base used. Adding more acid makes it more acidic, not safer. Common salt does not neutralise an acid. Sugar has no neutralising action and only dissolves.

10. (B) Turmeric turns red only in bases; staying yellow tells you the sample is not basic, but it cannot distinguish between an acid and a neutral substance, both of which leave turmeric yellow. So you cannot say it is definitely an acid or definitely neutral. It is certainly not a base, which rules out that option.

11. (B) Tamarind water is acidic (tartaric acid), lime water is basic, and sugar solution is neutral, matching acid then base then neutral. In the soap, lemon, NaOH list the order is base, acid, base, which is wrong. Distilled water, vinegar, milk of magnesia is neutral, acid, base, the wrong order. Baking soda, curd, salt water is base, acid, neutral, which has the acid and base swapped.

12. (C) Slaked lime is a base, so it neutralises the acidic soil, but added to already-basic pond water it would make it more basic, which is the contrast asked for. Vinegar is acidic,

not neutral, so the claim about it is wrong. Lemon juice is an acid, not a base. Sugar is neutral and does not neutralise either.

13. (A) Red litmus to blue means the solution is basic; bases turn turmeric red, so that result must follow. A base does not turn blue litmus red, since that is an acid reaction. A base turns phenolphthalein pink, not colourless. China rose turns green in bases, not dark pink, so that option is wrong.

14. (B) While dry, the acid and the base in baking powder sit side by side without reacting; only when water is added to the batter can they meet, react and release the gas that makes the cake rise. The acid does not lose its sourness in storage. Water is neutral, not a base that "switches on" the powder. The gas comes from the reaction, not from air beaten into the dry powder.

15. (C) Phenolphthalein is pink in base and colourless when not basic, so the pink vanishing means the added acid has neutralised the base, leaving the solution roughly neutral. It is not now strongly basic, since the indicator would still be pink. Lemon juice stays an acid. The soap solution is being neutralised, not made more basic.

16. (D) Phenolphthalein is a synthetic, lab-made indicator, so it is the one not from a living source. Litmus comes from lichens, turmeric from the turmeric plant, and china rose from hibiscus flowers, so all three are natural indicators.

17. (D) Vinegar is an acid and baking soda is a base; mixed in the sealed bag they neutralise and release a gas, and that trapped gas inflates the bag until it is taut. The bag does not swell from vinegar vapour, since there is no strong heating. No air enters a sealed bag from outside. Plain dissolving would not generate the gas needed to inflate the bag.

18. (C) Marble is calcium carbonate, which behaves as a base, so it reacts with the acidic rain and is gradually corroded. Marble is not acidic, so the first wrong option reverses the chemistry. Acid rain is acidic, not neutral, and it does more than wash dust off. Heating and cracking is not the cause of acid-rain damage.

19. (B) A wasp sting is basic, so it needs a mild acid like vinegar to neutralise it, which is the reverse of the bee or ant sting remedy. Baking soda is a base and would not neutralise a basic sting. Plain water only dilutes. Sodium hydroxide is a strong base and would worsen the sting, besides being dangerous on skin.

20. (A) Phenolphthalein turns pink only in a base. X turns turmeric red, the mark of a base, so X turns phenolphthalein pink. Y turns blue litmus red, so Y is an acid and gives no pink. Z changes neither litmus, so it is neutral and gives no pink. Phenolphthalein needs a base, not an acid, so the option claiming an acid turns it pink is wrong.

21. (A) An antacid is a base that neutralises stomach acid, producing salt and water, exactly fitting acid plus base gives salt plus water. Ice melting is a physical change of state.

An apple browning is a reaction with air, not an acid-base neutralisation. Salt dissolving is mixing, not a chemical reaction between an acid and a base.

22. (C) Blue litmus to red means acidic, and phenolphthalein is colourless in acids, so the two results agree and the sample is an acid. Phenolphthalein being involved does not make it basic; it is colourless here precisely because there is no base. Phenolphthalein is colourless in both acids and neutrals, but blue litmus turning red rules out neutral. The results are consistent, not contradictory.

23. (D) A neutral substance changes neither litmus paper, and water does exactly that, which is the evidence of neutrality. Turning blue litmus red is acid behaviour, which water does not show. Turning red litmus blue is base behaviour, also not shown by water. Phenolphthalein turning pink would mean a base, so that too would contradict neutrality.

24. (A) Calcium carbonate behaves as a base, so the eggshell reacts with the acid in vinegar, releasing a gas (the bubbles) while the shell is gradually eaten away. The vinegar does not dry the shell out — it is a liquid that reacts with it. The shell does not dissolve unchanged like sugar; a chemical reaction consumes it. There is no warming source to make the egg heat and flake the shell.

25. (C) Soap solution is basic, and methyl orange is yellow in bases, so the colour is yellow. Red would mean acid, but soap is not acidic. Saying red for a basic solution mismatches the rule, since red goes with acids. Soap is not neutral, and methyl orange is not colourless, so that option fails too.

26. (D) Different acid-base pairs give different salts; HCl with NaOH gives sodium chloride, while sulphuric acid with NaOH gives sodium sulphate, so the salt depends on the reactants. It is not always common salt. The product is a new salt, not the starting acid. Sodium hydroxide is the base used up, not the salt produced.

27. (B) Carbon dioxide acts as an acidic gas; once it just neutralises the basic lime water, the result is roughly neutral and leaves litmus unchanged. It is no longer strongly basic, so red litmus would not turn blue. Just neutralising does not make it strongly acidic, so blue litmus to red is wrong. Green is not a litmus colour.

28. (C) Turmeric stays yellow in acids and turns red in bases, which is correct. Phenolphthalein is the opposite of the stated wrong claim: colourless in acids and pink in bases. Blue litmus turns red in an acid, not a base. China rose is dark pink in acids and green in bases, the reverse of the wrong option.

29. (D) China rose turns green in a base; a base turns red litmus blue and feels soapy, so that is the matching set. Turning blue litmus red and tasting sour are acid properties, the opposite of a base. A base does change litmus, so leaving both unchanged is wrong. A base turns turmeric red, not staying yellow.

30. (A) Neutralisation between an acid and a base releases heat, so the mixture warms up. Evaporation would cool, not warm, and is not what is happening. Bases do not freeze a mixture. The warmth comes from the reaction itself, not from external sunlight.

31. (C) Lime water and ammonia solution are bases, so the count is two. Curd is acidic (lactic acid) and tamarind water is acidic (tartaric acid), so they are not bases. Thus one is too few and three or four is too many.

32. (B) Feeling soapy and turning turmeric red are both base properties, so the liquid is a base. An acid would feel sour-tasting (not soapy) and leave turmeric yellow. A neutral substance changes no indicator. Common salt solution is neutral and would not turn turmeric red.

33. (D) Slaked lime is a cheap, widely used base, so it neutralises the acidic water. Vinegar and lemon juice are acids, so they would worsen the acidity, not fix it. Adding more acidic runoff obviously makes the problem worse.

34. (D) Sour taste is the signature of acids, however weak or strong, so a stronger sour taste still means an acid, not a base. Strength of taste does not flip an acid into a base. Sourness never becomes basic. And sour does not mean neutral; neutral substances have no acid or base taste.

35. (D) Turmeric turns red only in a base. Q and S turn red, so two are basic. P and R stay yellow, so they are not basic (acid or neutral). Hence one is too few and three or four too many.

36. (C) Milk of magnesia is a base (magnesium hydroxide), so calling it an acid is the wrong pairing. Vinegar is correctly an acid, lime water is correctly a base, and curd is correctly an acid, so those three pairings are right.

37. (C) An antacid is a mild base, and it neutralises the excess HCl, forming salt and water and relieving the burning. Adding more acid would worsen the problem. Sugar coating does not neutralise the acid. Making the acid stronger is the opposite of relief.

38. (A) Phenolphthalein pink means solution 1 is a base. Solution 2 is colourless in phenolphthalein and turns blue litmus red, which marks it as an acid. So the order is base then acid. The reverse order is wrong. Solution 2 cannot be neutral because it turns blue litmus red. They are not both bases, since solution 2 is an acid.

39. (D) Baking soda solution is a base, so it turns red litmus blue and phenolphthalein pink. Lemon juice is an acid, not a base, so it does neither. Acids do not turn red litmus blue; that is a base reaction. Both are not acids, since baking soda is a base.

40. (B) Carbon dioxide dissolving makes carbonic acid, so a fizzy drink is mildly acidic. It is not basic, since an acid forms, not a base. It is not strongly basic for the same reason.

And it is not neutral, because the dissolved acid makes it acidic.

41. (D) A base must turn phenolphthalein pink, turmeric red, and china rose green, and it must turn red litmus blue. If it leaves litmus unchanged and phenolphthalein stays colourless, both point to it not being a base, so that disproves the claim. Phenolphthalein turning pink, turmeric turning red, or china rose turning green would each support, not disprove, the base claim.

42. (B) Lime improves acidic soil by neutralising it and worsens basic soil by adding more base, so lime is a base. It is not an acid; an acid would fix the basic soil, not the acidic one. It is not neutral, since it clearly changes the soil. It is not an inert salt, because it does change the soil character.

43. (C) At the endpoint the acid and base have neutralised, so the mixture is neutral and leaves both litmus papers unchanged, which is the confirmation. Blue litmus turning red would mean leftover acid, meaning the endpoint was overshoot with too much acid. Red litmus turning blue would mean leftover base. A neutral mixture is not a fresh acid.

44. (B) Turmeric turns red only in a base, and soap solution is basic, so it turns turmeric red. Lemon juice and vinegar are acids, which leave turmeric yellow. Sugar water is neutral and also leaves turmeric yellow.

45. (B) China rose is natural (from hibiscus) and phenolphthalein is synthetic, so that pair is one of each. Litmus and turmeric are both natural. Turmeric and china rose are both natural. Litmus and china rose are both natural, so only the first pair mixes natural with synthetic.

46. (D) A bee sting is acidic, so it needs a mild base such as baking soda to neutralise it, and honey is not a base. Honey is not basic, so it does not neutralise. Sweetness has nothing to do with neutralising an acid. Sweet substances do not neutralise acids, so that general claim is false.

47. (C) The first test shows an acid; after adding the base milk of magnesia, red litmus turning blue shows the mixture is now basic, meaning so much base was added that it overshoot neutral. Milk of magnesia is a base, not an acid. The original solution turned blue litmus red, so it was an acid, not a base. A clear change from acid to base proves a reaction did happen.

48. (A) Distilled water is neutral, and phenolphthalein stays colourless except in a base, so there is no colour. Distilled water is not acidic, and phenolphthalein does not turn pink in acids anyway. It is not basic, so the weak-base reason is wrong. Phenolphthalein responds to bases, not specifically to salt water.

49. (B) Blue litmus to red, phenolphthalein staying colourless, and china rose turning dark pink are all acid behaviours, so the contents are an acid and the results agree. China

rose changing to dark pink is the acid colour, not a base sign. Phenolphthalein colourless alone could be neutral, but the litmus change rules that out. The results are consistent, not contradictory.

50. (B) Common salt solution is neutral, so no indicator shows an acid or base colour: litmus unchanged, turmeric yellow, china rose unchanged, phenolphthalein colourless. The other sets mix acid signs (blue litmus red, turmeric staying yellow with acid) and base signs (turmeric red, phenolphthalein pink), none of which a neutral salt solution produces.

51. (C) Marble and limestone are calcium carbonate, a base that reacts with acid rain and is corroded. Glass is not acidic and is not the classic acid-rain casualty. Iron is a metal, not a base in this sense, though it rusts by a different process. Plastic does not neutralise acid and is not the example meant.

52. (A) Blue litmus turning red marks an acid. Vinegar and lemon juice turn it red, so two acids. Lime water is basic and milk is roughly neutral, so they do not turn blue litmus red. Hence two, not one, three, or four.

53. (D) Phenolphthalein is colourless in acids and in neutrals alike, so a colourless reading cannot distinguish them and a second indicator like litmus is needed. Litmus does not give the same colour in acids and bases; it clearly differs. Turmeric turns red only in bases, not in acids. Indicators do not all give one colour everywhere; that is why they are useful.

54. (A) Souring of milk is due to lactic acid forming, so spoiled milk is acidic. A base would taste bitter and not sour, so the base option is wrong. Becoming sour is moving away from neutral, not toward it. Sourness comes from an acid, not from a salt.

55. (D) After exact neutralisation the salt is dissolved in the water formed, so evaporating the water leaves the solid salt. Adding more acid or base would unbalance the neutral mixture and leave excess reactant, not cleanly recover the salt. Freezing would not separate out an acid, since no acid is left over.

56. (C) Red litmus turning blue means a base, and soap solution is basic, so it fits. Orange juice is acidic (citric acid). Vinegar is acidic (acetic acid). Soda water is acidic (carbonic acid). Only soap solution among these is a base.

57. (A) Fizzing stops and litmus is unchanged when the acidic lemon juice and basic baking soda have neutralised, giving a neutral mixture. It is not still strongly acidic, since litmus would then change. It is not strongly basic for the same reason. Neutralisation gives salt and water, not a stronger pure acid.

58. (B) Bitter taste, soapy feel, red litmus turning blue, and phenolphthalein turning pink are all base properties, so the liquid is clearly a base. A soapy feel is a base trait, not an acid

one. A neutral substance would not change any indicator, so many positive base results rule out neutral. The four results agree rather than conflict.

59. (A) Lime is a base, useful only for acidic soil; basic soil needs a mild acid such as decaying compost to neutralise it. Lime is not neutral, so it cannot fix any soil. Lime is a base, not an acid, so it would worsen basic soil. Sodium hydroxide is a strong base and would make basic soil even more basic, so it is the wrong fix.

60. (A) China rose turns green in a base, so cup B is basic, and phenolphthalein turns pink in a base, so cup B turns pink. Cup A is dark pink in china rose, which is the acid colour, so cup A is an acid and stays colourless with phenolphthalein. The two indicators agree rather than disagree, since both identify cup B as the base.